

GFIS Mid-Semester Presentation Outline

Recommended duration: 10 to 12 minutes, plus demo and questions.

Slide 1: Title

Green Fuel Intelligence System (GFIS): A Physics-Guided Hybrid AI Framework for Methane Yield Prediction, Soft Sensing and Digital-Twin-Ready Simulation in Anaerobic Digestion

Speak: Good morning respected evaluator and faculty members. My dissertation project is GFIS, a physics-guided AI framework for anaerobic digestion process intelligence.

Slide 2: Why This Problem Matters

- Anaerobic digestion converts organic waste into methane-rich biogas.
- Plant output depends on nonlinear biological, chemical and operating conditions.
- Small and medium plants often lack continuous laboratory-quality monitoring.
- Black-box prediction alone is risky because it may produce physically impossible or operationally unsafe outputs.

Visual: biogas process flow or GFIS front-page plant/control-room screenshot.

Slide 3: Abstract Feedback and Mid-Sem Response

- Abstract accepted with Excellent evaluation.
- Feedback pushed the work away from web prediction toward Physics-Informed Machine Learning, soft sensing and digital-twin-ready simulation.
- Mid-sem response: added explicit constraints, VFA/ALK soft sensor, stability classes, simulation engine and AWS deployment.

Use safe credit: The evaluator strengthened the project framing around Physics-Informed Machine Learning and digital-twin readiness during abstract evaluation.

Slide 4: Research Objective

Primary objective: Predict methane yield and process stability from anaerobic digestion variables using a hybrid AI model constrained by process feasibility.

Sub-objectives: - Build data pipeline for temperature, pH, OLR, HRT, TS, VS, C/N, moisture and ambient variables. - Train XGBoost and LSTM branches. - Add volatile-solids-based methane feasibility bound. - Estimate VFA/ALK as a soft sensor. - Provide scenario simulation and deployed demonstration interfaces.

Slide 5: System Architecture

Layers: - Data layer: raw data, preprocessing, lag/rolling features, time windows.
- Model layer: XGBoost, LSTM, ensemble, physics-guided audit. - Soft-sensor layer: VFA/ALK estimation and stability state. - Interface layer: FastAPI, Streamlit, Level 2 portal, control room, industrial simulator. - Deployment layer: AWS CloudFront web portal, AWS App Runner dashboard/API.

Slide 6: Dataset Used at Mid-Semester

Current dataset: - 2400 hourly synthetic/literature-inspired records. - Variables: timestamp, temperature, pH, OLR, HRT, TS, VS, C/N ratio, ambient temperature, moisture, methane yield, VFA/ALK ratio, stability label. - 250-row soft-sensor extract prepared for presentation and inspection.

Important honesty: This is controlled mid-sem validation data. Final validation will use real IoT/industry data and public datasets wherever available.

Slide 7: Mathematical Model

Input vector:

$x_i = [T_i, pH_i, OLR_i, HRT_i, TS_i, VS_i, C/N_i, moisture_i, ambient_i, lagged\ states_i]$

GFIS loss:

$L_{GFIS} = L_{CH4} + \lambda_{phys} L_{feas} + \lambda_{soft} L_{VFA/ALK} + \lambda_{stab} L_{stab} + \lambda_{re}$

Explain simply: The model is trained for accuracy, but it is also penalized when methane prediction violates physical feasibility or when soft-sensor/stability outputs are wrong.

Slide 8: Physics Constraint and Example

Methane feasibility bound:

$Y_{max_i} = VS_i \times BMP_{feedstock(i)} \times \eta_T(T_i)$

Penalty:

$L_{feas} = \text{mean}(\max(0, \hat{y}_i - Y_{max_i})^2 + \max(0, -\hat{y}_i)^2)$

Message: This prevents impossible predictions such as methane yield exceeding substrate-driven biochemical potential.

Slide 9: VFA/ALK Soft Sensor and Stability

Soft-sensor target:

$\hat{r}_i = g_{\phi}(x_i)$

Interpretable surrogate:

$$r_hat_i = \exp(\beta_0 + \beta_1 OLR_i + \beta_2 VS_i/TS_i + \beta_3 HRT_i^{-1} + \beta_4 |T_i - T_{opt}| + \beta_5 \max(0, pH_{opt} - pH_i))$$

Stability rule: - Stable: VFA/ALK < 0.30 - Warning: 0.30 <= VFA/ALK < 0.40 - Critical: VFA/ALK >= 0.40

Slide 10: Current Results

Current evaluation on 2400 records:

Model	R2	RMSE	MAE
LSTM	0.2942	11.5712	9.0685
XGBoost	0.7556	6.7933	5.5517
Ensemble	0.7554	6.8114	5.5699
Physics-guided ensemble	0.7554	6.8114	5.5699

Physics violation count: 0.

Interpretation: XGBoost is currently strongest because the data is tabular. LSTM is retained for real sequential data in the final phase.

Slide 11: Why Control Room and Simulator Are More Than Streamlit

Streamlit is one research interface. The full GFIS demonstration also includes: - Level 2 web portal for project navigation. - AI model control room for predictions and model interaction. - Industrial simulator with plant-like telemetry, live charts, warnings and scenario logic. - FastAPI backend for model endpoints. - AWS deployment for external access.

Say: The visible dashboard is the operator interface. The research contribution is the pipeline behind it: data, model, constraints, soft sensor, simulation and deployment.

Slide 12: Digital Twin Positioning

Current status: Digital-twin-ready simulation dashboard and process-intelligence prototype.

Not yet claimed: Fully synchronized industrial digital twin.

Reason: A true digital twin requires continuous physical plant synchronization through IoT/SCADA, which is planned for the next phase.

Slide 13: Live Demonstration

Demo order: 1. Open GFIS Level 2. 2. Open AI Model Control Room. 3. Show prediction/stability model interface. 4. Open Industrial Simulation Engine. 5. Run simulation and explain live telemetry. 6. Open Streamlit and FastAPI docs if time permits.

Use AWS links from `07_Demo_Links_For_Report.md`.

Slide 14: Next Phase

- Acquire plant IoT/industry data.
- Integrate public anaerobic digestion datasets.
- Re-train and calibrate soft sensor.
- Add uncertainty estimation and SHAP explanations.
- Strengthen LSTM with real time-series windows.
- Move from simulation prototype toward plant-synchronized digital twin.

Slide 15: Closing

GFIS has moved from accepted abstract to a working AWS-deployed research prototype. The main contribution is combining methane prediction, physics-guided feasibility, VFA/ALK soft sensing, stability interpretation and simulation-ready deployment for anaerobic digestion.

GFIS Mid-Semester Presentation Script

Opening

Good morning respected evaluator and faculty members. My dissertation project is titled Green Fuel Intelligence System, or GFIS. The project focuses on methane yield prediction, soft sensing and digital-twin-ready simulation for anaerobic digestion using a physics-guided hybrid artificial intelligence framework.

The abstract stage was accepted with an Excellent evaluation. The feedback helped refine the project direction away from a simple web prediction system and toward Physics-Informed Machine Learning, process soft sensing and digital-twin-ready simulation. In this mid-semester version, I have implemented that direction through model training, feasibility constraints, VFA/ALK soft sensing, stability classification, scenario simulation and AWS-deployed interfaces.

Problem Statement

Anaerobic digestion is a nonlinear biological and biochemical process. Methane production depends on temperature, pH, organic loading rate, hydraulic retention time, total solids, volatile solids, C/N ratio, moisture, ambient temperature and past reactor state. A normal machine-learning model can learn useful patterns,

but it does not automatically know physical feasibility. For example, it may predict a methane value that is higher than what the available volatile solids can realistically produce.

Therefore, GFIS is designed to answer three questions. First, what methane yield is expected? Second, is the reactor condition Stable, Warning or Critical? Third, is the predicted methane yield physically plausible?

Current System

The current system has five layers.

The first layer is the dataset and feature pipeline. It prepares anaerobic digestion records containing temperature, pH, OLR, HRT, TS, VS, C/N ratio, ambient temperature, moisture, methane yield, VFA/ALK ratio and stability label. It also supports lag features, rolling features and time windows for sequential modelling.

The second layer is the machine-learning layer. It includes XGBoost for tabular prediction, PyTorch LSTM for time-series modelling, an ensemble branch, and a physics-guided feasibility audit.

The third layer is the soft-sensor layer. This estimates VFA/ALK ratio from easier-to-measure operating variables. The reason is that VFA/ALK is important for stability, but in small plants it may not be measured continuously.

The fourth layer is the simulation layer. It lets us run controlled scenarios such as organic overload, pH shock, low temperature disturbance and HRT washout risk.

The fifth layer is the deployment layer. The system is available through a Level 2 web portal, model control room, industrial simulation engine, Streamlit research dashboard and FastAPI backend deployed on AWS.

Mathematical Explanation

The input vector is:

$x_i = [T_i, pH_i, OLR_i, HRT_i, TS_i, VS_i, C/N_i, moisture_i, ambient_i, lagged\ states_i]$

The main loss function is:

$L_{GFIS} = L_{CH4} + \lambda_{phys} L_{feas} + \lambda_{soft} L_{VFA/ALK} + \lambda_{stab} L_{stab} + \lambda_{reg}$

Here, L_{CH4} is the methane prediction error. L_{feas} is the physics feasibility penalty. $L_{VFA/ALK}$ is the soft-sensor error. L_{stab} is the stability classification error. The regularization term prevents overfitting.

The methane prediction error is:

$L_{CH4} = \text{mean}((y_i - \hat{y}_i)^2)$

The physics penalty is:

$$L_{\text{feas}} = \text{mean}(\max(0, y_{\text{hat}_i} - Y_{\text{max}_i})^2 + \max(0, -y_{\text{hat}_i})^2)$$

This means if the predicted methane is above the physical upper bound, the model is penalized. If the predicted methane is negative, it is also penalized. So the model is not allowed to behave like a completely unconstrained black box.

The feasibility bound is:

$$Y_{\text{max}_i} = \text{VS}_i \times \text{BMP_feedstock}(i) \times \text{eta_T}(T_i)$$

This is a volatile-solids-based upper bound. It says methane production is limited by available biodegradable volatile solids, the biochemical methane potential of the feedstock, and the temperature efficiency factor.

Example Calculation

Assume:

$$\text{VS} = 7.2 \text{ kg}$$

$$\text{BMP} = 28 \text{ m}^3 \text{ CH}_4 \text{ per kg VS-equivalent scaling unit}$$

$$\text{eta_T} = 0.92$$

Then:

$$Y_{\text{max}} = 7.2 \times 28 \times 0.92 = 185.47 \text{ m}^3/\text{day}$$

If the model predicts:

$$y_{\text{hat}} = 190 \text{ m}^3/\text{day}$$

Then the feasibility violation is:

$$190 - 185.47 = 4.53$$

$$L_{\text{feas contribution}} = 4.53^2 = 20.52$$

If the model predicts:

$$y_{\text{hat}} = 175 \text{ m}^3/\text{day}$$

Then it is below the feasibility bound, so the feasibility penalty is zero.

This is the core idea of GFIS: prediction plus physical checking.

Current Results

At mid-semester, the dataset contains 2400 hourly records. The current XGBoost model performs best with R2 of 0.7556, RMSE of 6.7933 and MAE of 5.5517. The physics-guided ensemble has zero physics-bound violations in the current evaluation.

The LSTM branch currently has lower R2, around 0.2942. I am not removing it because LSTM becomes more meaningful when actual sequential plant data is

available. The current result tells us that for this mid-semester tabular dataset, XGBoost is stronger. For the final phase, LSTM will be retrained with real time-series data and lagged reactor histories.

Digital Twin Clarification

I am deliberately calling the current system a digital-twin-ready simulation dashboard, not a full industrial digital twin. A true digital twin must be continuously synchronized with a physical plant through IoT or SCADA data. At present, GFIS demonstrates the software foundation: data pipeline, model, constraints, soft sensor, scenario engine, API and dashboards. Once live sensor data is connected, this architecture can mature into a plant-synchronized digital twin.

Demonstration Script

For demonstration, I will first open the GFIS Level 2 research workbench. This is the central portal for the dissertation prototype.

Then I will open the AI Model Control Room. This interface shows the model-facing side: methane prediction, stability state, API-backed logic and evidence links.

Next I will open the Industrial Simulation Engine. This page is not only visual decoration. It is an accelerated simulation interface that updates plant variables, graphs, logs and warning states. It demonstrates how an operator might inspect methane output, stability and process disturbances.

After that, if time permits, I will show the Streamlit research dashboard and FastAPI documentation. Streamlit is useful for research workflow, while FastAPI shows that the system is structured as backend services and not just static HTML.

Closing

In summary, GFIS has progressed from an accepted abstract to a working AWS-deployed mid-semester prototype. The key contribution is the combination of methane prediction, physics-guided feasibility checking, VFA/ALK soft sensing, stability classification and simulation-ready deployment for anaerobic digestion. The next phase will focus on stronger datasets, real IoT/industry data, explainability, uncertainty and true digital twin synchronization.

GFIS Mathematical Equations and Working Calculations

1. Input Feature Vector

For each time step or sample i , GFIS uses measurable anaerobic digestion variables:

$x_i = [T_i, pH_i, OLR_i, HRT_i, TS_i, VS_i, C/N_i, \text{moisture}_i, \text{ambient}_i, \text{lagged reactor states}_i]$

Where:

Symbol	Meaning
T_i	Reactor temperature
pH_i	Reactor pH
OLR_i	Organic loading rate
HRT_i	Hydraulic retention time
TS_i	Total solids
VS_i	Volatile solids
C/N_i	Carbon-to-nitrogen ratio
moisture_i	Feedstock moisture
ambient_i	Ambient temperature

2. Methane Prediction Model

The model predicts methane yield:

$y_{\text{hat}_i} = f_{\text{theta}}(x_i)$

where f_{theta} can be XGBoost, LSTM or an ensemble.

Basic methane error:

$L_{\text{CH4}} = (1/N) \sum_i (y_i - y_{\text{hat}_i})^2$

3. Physics-Guided GFIS Objective

The full research objective is:

$$L_{\text{GFIS}} = L_{\text{CH4}} + \lambda_{\text{phys}} L_{\text{feas}} + \lambda_{\text{soft}} L_{\text{VFA/ALK}} + \lambda_{\text{stab}} L_{\text{stab}} + \lambda_{\text{reg}} ||\theta||_2^2$$

Interpretation:

Term	Role
L_{CH4}	Predict methane accurately
L_{feas}	Penalize physically impossible methane outputs

Term	Role
L_VFA/ALK	Train soft sensor for VFA/ALK ratio
L_stab	Train Stable/Warning/Critical classifier
lambda terms	Control the importance of each objective

4. Methane Feasibility Bound

Current mid-semester bound:

$$Y_{\max_i} = VS_i \times BMP_{\text{feedstock}(i)} \times \eta_T(T_i)$$

Expanded future bound:

$$Y_{\max_i} = VS_i \times BMP_{\text{feedstock}(i)} \times \eta_T(T_i) \times \eta_{\text{pH}}(\text{pH}_i) \times \eta_{\text{HRT}}(\text{HRT}_i)$$

Meaning:

Symbol	Meaning
$Y_{\max_i}$	Maximum feasible methane under sample condition
VS_i	Volatile solids available for conversion
$BMP_{\text{feedstock}}$	Biochemical methane potential of feedstock
η_T	Temperature efficiency factor
η_{pH}	pH efficiency factor
η_{HRT}	Retention-time efficiency factor

5. Feasibility Penalty

$$L_{\text{feas}} = (1/N) \sum_i [\max(0, \hat{y}_i - Y_{\max_i})^2 + \max(0, -\hat{y}_i)^2]$$

This catches two impossible cases:

1. Predicted methane exceeds the physical upper bound.
2. Predicted methane is negative.

6. Worked Example: No Violation

Assume:

$$VS = 7.20$$

$$BMP_{\text{feedstock}} = 28.00$$

$$\eta_T = 0.92$$

$$\hat{y} = 175.00$$

Calculation:

$$Y_{\max} = 7.20 \times 28.00 \times 0.92$$

$$Y_{\max} = 185.47$$

Since:

$$y_{\text{hat}} = 175.00 < 185.47$$

Penalty:

$$\max(0, 175.00 - 185.47)^2 = 0$$

Result: prediction is feasible.

7. Worked Example: Physics Violation

Assume:

$$VS = 7.20$$

$$BMP_{\text{feedstock}} = 28.00$$

$$\eta_T = 0.92$$

$$y_{\text{hat}} = 190.00$$

Calculation:

$$Y_{\text{max}} = 185.47$$

$$\text{violation} = 190.00 - 185.47 = 4.53$$

$$\text{penalty} = 4.53^2 = 20.52$$

Result: the prediction is penalized because it exceeds the substrate-driven methane feasibility limit.

8. VFA/ALK Soft Sensor

General form:

$$r_{\text{hat}_i} = g_{\text{phi}}(x_i)$$

Interpretable engineering surrogate:

$$\begin{aligned} r_{\text{hat}_i} = \exp(&\beta_0 + \beta_1 \text{OLR}_i + \beta_2 (VS_i / TS_i) \\ &+ \beta_3 \text{HRT}_i^{-1} + \beta_4 |T_i - T_{\text{opt}}| \\ &+ \beta_5 \max(0, \text{pH}_{\text{opt}} - \text{pH}_i)) \end{aligned}$$

Interpretation:

- Higher OLR can increase acid loading.
- Higher VS/TS can indicate stronger biodegradable organic fraction.
- Lower HRT can increase washout/instability risk.
- Temperature deviation can disturb microbial activity.
- Low pH can indicate acid stress.

9. Stability Classification

Current rule:

Stable if $r_{\text{hat}} < 0.30$
Warning if $0.30 \leq r_{\text{hat}} < 0.40$
Critical if $r_{\text{hat}} \geq 0.40$

Worked examples:

VFA/ALK estimate	Stability
0.24	Stable
0.34	Warning
0.45	Critical

10. How To Explain In Viva

Say:

GFIS is not only optimizing model accuracy. It adds process knowledge. The model can learn nonlinear relations from data, but methane output is checked against a volatile-solids-based feasibility bound. This makes the system more suitable for biological process intelligence than a plain regression dashboard.

GFIS Dashboards, Control Room and Industrial Simulator Explanation

Short Answer

No, the GFIS demonstration is not only Streamlit.

Streamlit is one research dashboard. The full GFIS system includes a deployed web portal, model control room, industrial simulation interface, FastAPI backend and AWS deployment.

Deployed Components

Component	Purpose	Live Link
GFIS Portal	Public project entry point and level navigation	https://d2c2mw7lwtswdx.cloudfront.net/
Level 2 Research Workbench	Main dissertation workbench	https://d2c2mw7lwtswdx.cloudfront.net/level-2/
AI Model Control Room	Model-facing UI for GFIS prediction and research workflow	https://d2c2mw7lwtswdx.cloudfront.net/level-2/apps/model-control-room/

Component	Purpose	Live Link
Industrial Simulation Engine	Plant-like telemetry and scenario simulation interface	https://d2c2mw7lwtswdx.cloudfront.net/level-2/apps/model-control-room/industrial_simulation.html
Streamlit Dashboard	Research dashboard for inspection and interaction	https://ea4js82me5.us-east-1.awsapprunner.com/
FastAPI Backend	API layer for prediction, simulation and docs	https://y3kgjnwetp.us-east-1.awsapprunner.com/docs

Why The Control Room Is More Than A Static Dashboard

A static dashboard only displays values. The GFIS control room and simulator demonstrate the process-intelligence pipeline:

1. Process variables are represented as reactor state inputs.
2. Methane prediction is linked to the model layer.
3. Physics feasibility is shown as a constraint/audit concept.
4. VFA/ALK soft-sensor output is mapped to stability state.
5. Scenario simulation shows how conditions change over a time horizon.
6. API and AWS deployment show that the prototype is structured for real service integration.

Industrial Simulation Engine

The industrial simulation engine should be explained as an accelerated virtual plant interface. It visually behaves like a plant control room: telemetry cards, live charts, terminal-style logs, warning states and scenario controls.

It demonstrates:

- Methane-production behavior over time.
- Reactor temperature and pH trends.
- VFA/ALK stability movement.
- Warning/critical states under disturbances.
- Operator-style interpretation of scenario behavior.

What It Is Not Yet

It is not yet a fully synchronized industrial digital twin because it is not connected continuously to a physical plant through IoT/SCADA.

Best Viva Wording

Use this wording:

The Streamlit dashboard is only one research interface. The GFIS Level 2 system also includes a model control room, an industrial simulation engine, FastAPI services and AWS deployment. The current industrial simulator is an accelerated digital-twin-ready simulation prototype. It demonstrates the modelling and operator workflow, while full digital-twin status will require continuous real plant data ingestion in the next phase.

Demonstration Order

1. Open Level 2 workbench.
2. Launch AI Model Control Room.
3. Show model/prediction/stability evidence.
4. Launch Industrial Simulation Engine.
5. Run a scenario and point to live telemetry.
6. Show Streamlit and FastAPI only if time remains.

GFIS Dataset and Open Data Strategy

Current Mid-Semester Dataset

The current mid-semester validation uses a controlled synthetic/literature-inspired anaerobic digestion dataset.

Files copied in this pack:

- 03_Datasets/GFIS_raw_synthetic_anaerobic_digestion.csv
- 03_Datasets/GFIS_processed_synthetic_anaerobic_digestion.csv
- 03_Datasets/GFIS_soft_sensor_dataset_extract.csv
- 03_Datasets/evaluation_metrics.json
- 03_Datasets/evaluation_report.md

Dataset shape:

File	Rows	Columns
Raw synthetic AD dataset	2400	13
Processed synthetic AD dataset	2400	13
Soft-sensor extract	250	13

Columns:

timestamp, temperature, pH, OLR, HRT, TS, VS, C_N_ratio,
ambient_temperature, moisture, methane_yield, VFA_ALK_ratio, stability_label

Why Synthetic/Literature-Inspired Data Is Used Now

The mid-semester objective is to prove that the pipeline works:

- data ingestion,
- feature handling,
- model training,
- physics feasibility audit,
- soft sensing,
- stability classification,
- API/dashboard interfaces,
- simulation workflow.

Final dissertation claims should be based on stronger data:

- real IoT data,
- plant/industry logs,
- laboratory data,
- public datasets,
- benchmark dynamic simulation datasets.

Public Dataset Options To Explore

These are useful leads for the next phase:

1. Kaggle: Biogas Production Analysis
<https://www.kaggle.com/datasets/dineshsharma132/biogas-production-analysis>
2. Kaggle: Indian Biogas Production Dataset
<https://www.kaggle.com/datasets/ucimachinelearning/indian-biogas-production-dataset>
3. Kaggle: Livestock Anaerobic Digester Database
<https://www.kaggle.com/datasets/mehmetisik/livestock-anaerobic-digester-database>
4. Mendeley Data: Process monitoring and weather datasets on farm-scale biogas systems
<https://data.mendeley.com/datasets/gk3f363sfg>
5. Data in Brief / ScienceDirect: Comprehensive anaerobic digestion dataset for cattle manure, source-separated organics, primary sludge and waste activated sludge
<https://www.sciencedirect.com/science/article/pii/S2352340919302641>

Suggested Dataset Plan For Final Dissertation

Layer	Purpose
Synthetic/literature-inspired data	Controlled pipeline testing
Public dataset	External validation and model comparison
IoT/plant data	Final real-world validation
ADM1 or benchmark simulation data	Mechanistic comparison and stress testing

What To Say In Presentation

Say:

At mid-semester, the results are preliminary and based on controlled synthetic/literature-inspired anaerobic digestion data. The purpose is to validate the end-to-end GFIS pipeline. In the final phase, I will integrate real IoT/industry readings and public datasets so that the system can move from prototype validation to stronger real-world validation.

Evaluator Credit and Response Wording

Safe Portal Wording

Use this on the portal:

The BITS abstract evaluation helped strengthen GFIS toward Physics-Informed Machine Learning, soft sensing and digital-twin-ready simulation. GFIS uses PIML in its established meaning of Physics-Informed Machine Learning, while acknowledging the evaluator's guidance for sharpening this research positioning during the abstract evaluation.

Safe Presentation Wording

Say:

During the abstract evaluation, the evaluator specifically encouraged the work to move away from black-box machine learning toward Physics-Informed Machine Learning, soft sensing and digital-twin-ready simulation. I have incorporated that direction in the mid-semester version by adding explicit physics constraints, VFA/ALK soft sensing, stability classification and deployed simulation interfaces.

Avoid Saying

Do not say:

- The evaluator invented PIML.
- The evaluator coined the entire field.
- The work is already a full industrial digital twin.
- The present dataset is real industrial data.

Correct Technical Position

Say:

- PIML is an established term: Physics-Informed Machine Learning.
- The evaluator strengthened the GFIS framing around PIML.
- The current system is a digital-twin-ready simulation prototype.
- Real-time IoT/SCADA integration is the next phase.

Response To Abstract Evaluation

Evaluation point	Response in mid-semester work
Move away from black-box ML	Added physics-guided objective and methane feasibility constraint
Predict hard-to-measure chemical states	Added VFA/ALK soft sensor
Avoid overclaiming digital twin	Uses digital-twin-ready simulation wording
Use real high-quality data	Planned IoT/industry data and public dataset integration
Show working system	AWS portal, control room, simulator, Streamlit and API are deployed

GFIS Demo Links For Report and Presentation

Use these AWS links instead of localhost.

Public Portal

GFIS Portal:
<https://d2c2mw7lwtswdx.cloudfront.net/>

Gamma Presentation

GFIS Mid-Semester Presentation:
<https://green-fuel-intelligence--mtmb0aw.gamma.site/>

Local exported PDF:

08_Final_Presentation_Assets/GFIS_MidSem_Presentation_Gamma_Export_2026-07-09.pdf

Dissertation Workbench

GFIS Level 2 Research Workbench:
<https://d2c2mw7lwtswdx.cloudfront.net/level-2/>

Model Control Room

AI Model Control Room:
<https://d2c2mw7lwtswdx.cloudfront.net/level-2/apps/model-control-room/>

Industrial Simulation

Industrial Digital-Twin-Ready Simulation Engine:
https://d2c2mw7lwtswdx.cloudfront.net/level-2/apps/model-control-room/industrial_simulation

Streamlit Research Dashboard

Streamlit Research Dashboard:
<https://ea4js82me5.us-east-1.awsapprunner.com/>

FastAPI Backend

GFIS API Documentation:
<https://y3kgjnwetp.us-east-1.awsapprunner.com/docs>

GFIS API Health Endpoint:
<https://y3kgjnwetp.us-east-1.awsapprunner.com/health>

Report Wording

Use:

The GFIS prototype is deployed on AWS. The static GFIS portal and Level 2 web interfaces are served through Amazon CloudFront, while the Streamlit research dashboard and FastAPI backend are deployed through AWS App Runner. This deployment allows the evaluator to inspect the prototype externally without relying on localhost services.

Possible Mid-Semester Questions and Answers

Q1. Is this a real digital twin?

Answer: At present it is a digital-twin-ready simulation dashboard and process-intelligence prototype. A true digital twin requires continuous synchronization with a physical plant through IoT or SCADA. GFIS currently builds the modelling, API, dashboard, soft-sensor and simulation foundation needed for that next phase.

Q2. Why use physics constraints if machine learning already predicts methane?

Answer: Prediction accuracy alone is not enough for a biochemical process. A model can extrapolate and produce physically impossible methane values. The physics constraint checks methane prediction against a volatile-solids-based feasibility bound, so the model behaves more like a process-intelligence tool than a black-box regression model.

Q3. Why is XGBoost better than LSTM at this stage?

Answer: The current mid-semester dataset is mostly tabular and synthetic/literature-inspired. XGBoost is strong for tabular nonlinear relationships. LSTM is retained because the final phase targets real sequential plant data, where retention-time effects and lagged reactor history should matter more.

Q4. What is the soft sensor?

Answer: The soft sensor estimates VFA/ALK ratio from easier-to-measure variables such as temperature, pH, OLR, HRT, TS and VS. VFA/ALK is important for stability but may not be continuously available in small plants. The soft sensor converts measurable operating data into an estimated chemical-risk state.

Q5. Why is the dataset synthetic?

Answer: The current dataset validates the pipeline under controlled conditions. It allows model training, physics constraint testing, simulation and interface development before real plant data is available. The final dissertation phase will add IoT/industry data and public datasets for stronger validation.

Q6. What is the biggest contribution at mid-semester?

Answer: The contribution is not just a dashboard. The mid-semester contribution is an end-to-end deployed research prototype combining methane prediction, physics feasibility checking, VFA/ALK soft sensing, stability classification, scenario simulation, API services and AWS deployment.

Q7. How will you validate the final version?

Answer: The final version will be validated using stronger datasets: public anaerobic digestion datasets, plant IoT readings if available, and benchmark or ADM1-style simulation data. I will compare XGBoost, LSTM, ensemble and physics-guided outputs using R2, RMSE, MAE, physics violation count, stability accuracy and scenario behavior.

GFIS Control Room to Industrial Simulator Coordinator

What Was Added

The GFIS Model Control Room and Industrial 48-hour Simulation Engine now use a shared scenario handoff.

The control room publishes the current process-variable state:

temperature, pH, OLR, HRT, TS, VS, C/N ratio, ambient temperature, moisture

The industrial simulator imports that state and uses it to modify:

- VFA/ALK soft-sensor pressure,
- methane yield trajectory,
- methane quality,
- warning/critical alarm behavior,
- 48-hour simulated process response.

Operator Workflow

1. Open the Model Control Room.
2. Adjust process variables such as OLR, pH, temperature, HRT, TS or VS.
3. Run prediction or OLR sweep.
4. Click **Send to Industrial Simulator**.
5. The simulator opens with the same process state.
6. Run the 48-hour simulation and observe how the plant behavior changes.

Why This Matters

Earlier, the simulator mainly showed accelerated process behavior over time. Now it also responds to operator-set process variables. This makes the demonstration closer to a digital-twin workflow:

operator input -> process-state package -> simulation response -> stability interpretation

What Changes In The Simulation

High-risk conditions increase VFA/ALK pressure:

- high OLR,
- low pH,
- low HRT,
- high solids,
- temperature deviation from mesophilic optimum.

The simulator then modifies methane output and warning behavior. This lets the evaluator see what changes when process variables are changed, instead of only watching a fixed animation.

Digital Twin Positioning

This is still not a full industrial digital twin, because it is not continuously connected to a physical plant through IoT/SCADA. The correct phrase is:

GFIS now demonstrates a digital-twin-ready coordination pattern, where control-room decisions are transferred into an industrial process simulator and reflected in the 48-hour simulated plant response.

Viva Explanation

Use this:

The model control room is the decision layer, and the industrial simulator is the virtual plant response layer. When I change OLR, pH or temperature in the control room, that state is handed to the simulator. The simulator recalculates VFA/ALK pressure and methane behavior during the 48-hour horizon. This demonstrates the digital-twin workflow we will later connect to real IoT or SCADA data.

GFIS Run Memory, Export System and Level 1 Government Proposal Workbench

What Was Added

GFIS now has local experiment memory in the control room and industrial simulator.

This solves an important research issue: if the simulator runs but does not store data, it is only a visual demo. With memory and export, each run becomes evidence that can be used for reports, model analysis and viva explanation.

Control Room Memory

The Model Control Room now records:

- prediction runs,
- parameter changes,
- OLR sweep runs,
- 48-hour plant traces,
- optimizer changes,
- handoff events sent to the industrial simulator.

It stores:

timestamp, action, temperature, pH, OLR, HRT, TS, VS,
methane_yield, VFA_ALK_ratio, stability_label, parameter effect

Export options:

- CSV,
- JSON,
- Markdown report,
- reset memory after export.

Industrial Simulator Memory

The industrial simulator now records:

- simulation start and pause,
- process-variable application,
- periodic 48-hour simulation samples,
- VFA/ALK pressure,
- methane yield,
- methane quality,
- current process state.

Export options:

- CSV,
- Markdown report,
- reset simulator memory after export.

Why This Matters For LSTM and XGBoost

The exported logs can become training and validation data for later modelling.

For XGBoost:

- each row can act as a tabular sample,
- features are process variables,
- targets are methane yield, VFA/ALK and stability label.

For LSTM:

- exported time-ordered simulator records can form sequence windows,
- reactor behavior over the 48-hour horizon can be treated as temporal data,
- future IoT records can replace the simulated records.

Level 1 Government Proposal Workbench

A new Level 1 local page was added:

http://127.0.0.1:8561/GFIS_Level1_Government_Proposal_Local/

It explains GFIS as a government-scale planning system:

- national plant distribution,
- district hub-and-spoke planning,

- village/taluka suitability scoring,
- K-Means style cluster planning,
- XGBoost/LSTM methane prediction,
- IoT monitoring,
- copyright/IP positioning.

Level 1 to Level 2 Story

Use this explanation:

Level 1 answers where biogas plants should be installed. It uses GIS, feedstock availability, infrastructure access and economic suitability for plant distribution planning. Level 2 answers how each plant should be monitored and optimized after installation. It uses PIML, XGBoost, LSTM, VFA/ALK soft sensing and 48-hour simulation for plant operation.

Copyright/IP Reference

The presentation pack now includes:

- GFIS Level 1 DIPEX report,
- GFIS copyright application,
- GFIS copyright acknowledgement slip,
- author NOC.

Folder:

`07_Level1_and_IP_Reference/`

Use careful wording:

GFIS is being treated as protected software/IP. The available copyright application and acknowledgement documents are included as evidence. Final legal claims should be reviewed by legal counsel before government or commercial submission.

New Screenshots

Screenshots added:

- `08_control_room_memory_export_panel.png`
- `09_industrial_simulator_memory_panel.png`
- `10_level1_government_planner_localhost.png`